MAT3007 Homework # 6 Spring 2024

Due: Friday, April 19 at 12:00pm (noon)

Please note: You should submit your answers (scanned or typeset) in a PDF file. The problem set is due at 12:00pm (noon) on Friday, April 19th.

1. (30 pts) Consider the following functions:

$$
\begin{aligned}
a(x) &= 2x_1^2 - 30x_1 + 10 \\
b(x) &= -4x_2^2 + 20x_2 \\
c(x) &= \ln(x_1 + 1) + \ln(x_2 + 2) \\
d(x) &= 5x_1^2 + 3x_2^2 \\
e(x) &= -2x_1^2 + x_1x_2 - 4x_2^2 \\
f(x) &= x_1 + x_2
\end{aligned}
$$

For each of the following, indicate whether the feasible region is a convex set, and classify each model as to whether it is a convex program:

(a) $\min\{a(x) - b(x) : f(x) \leq 10\}$

(b) $\max\{b(x) + c(x) : d(x) \leq 10; x \geq 0\}$

(c) $\max\{d(x) : c(x) \geq 4, e(x) \geq -5, f(x) \leq 10, x \geq 0\}$

(d) $\min\{d(x) : c(x) \geq 4, e(x) \geq -5, f(x) \leq 10, x \geq 0\}$

Solution 1.

(a) $f(x) \leq 10$ is a convex constraint, so the feasible region is a convex set.
$\nabla^2[a(x) - b(x)] = \begin{bmatrix} 4 & 0 \\ 0 & 8 \end{bmatrix}$ is positive definite, so this is a convex function.
Minimize a convex problem over a convex feasible region, so it's a convex program.

(b) $d(x) \leq 10$ and $x \geq 0$ are both convex constraints, so the feasible region is a convex set.
$\nabla^2[b(x) + c(x)] = \begin{bmatrix} -\frac{1}{(x_1+2)^2} & 0 \\ 0 & -8 - \frac{1}{(x_2+2)^2} \end{bmatrix}$ is negative definite, so this is a concave function.
Maximize a concave function over a convex feasible region, so it's a convex program.

(c) $c(x) \geq 4, e(x) \geq -5, f(x) \leq 10, x \geq 0$ are all convex constraints, so the feasible region is a convex set.
$\nabla^2[d(x)] = \begin{bmatrix} 10 & 0 \\ 0 & 6 \end{bmatrix}$ is positive definite, so this is a convex function.
Maximize a convex function over a convex feasible region, so it's not a convex program.

(d) According to (c), the feasible region is a convex set, and this is a convex program.

2. (30 pts) Is the following function convex?

$$f(x) = 5x_1 + 2x_2^2 + x_3^2 - 3x_3x_4 + 4x_4^2 + 2x_5^4 + x_5^2 + 3x_5x_6 + 6x_6^2 + 3x_6x_7 + x_7^2$$

Hint: Try to answer without building a Hessian that is any larger than need be.

Solution 2.

$$\begin{aligned} f(x) &= 5x_1 + 2x_2^2 + x_3^2 - 3x_3x_4 + 4x_4^2 + 2x_5^4 + x_5^2 + 3x_5x_6 + 6x_6^2 + 3x_6x_7 + x_7^2 \\ &= 5x_1 + 2x_2^2 + \left(x_3^2 - 3x_3x_4 + 4x_4^2\right) + 2x_5^4 + \left(x_5^2 + 3x_5x_6 + 3x_6^2\right) + \left(3x_6^2 + 3x_6x_7 + x_7^2\right) \end{aligned}$$

$g_1(x) = 5x_1 + 2x_2^2$ is a convex function.

$g_2(x) = x_3^2 - 3x_3x_4 + 4x_4^2, \nabla^2 g_2(x) = \begin{bmatrix} 2 & -3 \\ -3 & 8 \end{bmatrix}$ is positive definite, so $g_2(x)$ is a convex function.

$g_3(x) = 2x_5^4$ is a convex function.

$g_4(x) = x_5^2 + 3x_5x_6 + 3x_6^2$, $\nabla^2 g_4(x) = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$ is positive definite, so $g_4(x)$ is a convex function.

$g_5(x) = 3x_6^2 + 3x_6x_7 + x_7^2$, $\nabla^2 g_5(x) = \begin{bmatrix} 6 & 3 \\ 3 & 2 \end{bmatrix}$ is positive definite, so $g_4(x)$ is a convex function.

Therefore, $f(x) = \sum_{i=1}^5 g_i(x)$ is also a convex function.

3. (40 pts) Let $m \in \mathbb{R}^n$ and $r > 0$ be given and define the ball $C := \left\{x \in \mathbb{R}^n : \|x - m\|_2^2 \leq r^2\right\}$, where $\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$ stands for the two-norm. We want to compute the projection of any point $x \in \mathbb{R}^n$ onto the ball, i.e., we want to solve the optimization problem:

$$\begin{aligned} &\min_{y \in \mathbb{R}^n} \frac{1}{2}\|y - x\|^2 \\ &\text{s.t. } \|y - m\|^2 \leq r^2 \end{aligned}$$

(a) Write down the KKT conditions for this projection problem.

(b) Show that the KKT conditions have a unique solution and calculate the corresponding KKT primal-dual pair explicitly.

Solution 3.

(a) The Lagrangian the problem is given by $L(y, \lambda) = \frac{1}{2}\|y - x\|^2 + \lambda\left(\|y - m\|^2 - r^2\right)$. Hence, the KKT conditions are given by:

$$\nabla_y L(y, \lambda) = y - x + 2\lambda(y - m) = 0, \quad \lambda \geq 0, \quad \lambda\left(\|y - m\|^2 - r^2\right) = 0, \quad \|y - m\| \leq r.$$

(b) We first assume that $\|y-m\| < r$. In this case, the complementarity conditions imply $\lambda = 0$ and using the main condition, we can infer $y = x$. (Thus, this case requires $\|x-m\| < r$). Next, suppose that $\|y-m\| = r$. Rearranging the main condition, it follows

$$y = m + \frac{x-m}{1+2\lambda} \quad \text{and} \quad r = \|y-m\| = \frac{\|x-m\|}{1+2\lambda} \quad \Longrightarrow \quad 2\lambda = \frac{\|x-m\|-r}{r}.$$

Consequently, we obtain $y = m + r \cdot \frac{x-m}{\|x-m\|}$. In addition, our calculations show $\|x-m\| = (1+2\lambda)r \geq r$.
Overall, we can summarize: In the case $\|x-m\| < r$, the unique KKT pair (y^*, λ^*) is given by $(y^*, \lambda^*) = (x, 0)$. In the case $\|x-m\| \geq r$, the unique KKT pair (y^*, λ^*) is given by $(y^*, \lambda^*) = \left(m + r\frac{x-m}{\|x-m\|}, \frac{1}{2}\left(\frac{\|x-m\|}{r} - 1\right)\right)$.